HANDS-ON ACTIVITY FOCUS GUIDE

# Predict, Command, Observe, Verify

Connect setPower() and telemetry to real wheel behavior

| 25 MIN<br>ACTIVITY | 8<br>STUDENT ROLES | CONTROLLED TEST<br>FORMAT | 1<br>SAFETY LEAD |
|---|---|---|---|

**ACTIVITY OUTCOME** Students predict motion from left/right power pairs, identify object-method-argument syntax, and use telemetry plus observation to diagnose one mismatch at a time.

## Materials

- Robot secured on stable blocks with both wheels clear.
- Prediction cards: 0.0/0.0, +0.3/+0.3, -0.3/-0.3, +0.3/-0.3, and +0.5/+0.2.
- Driver Station showing Left Power and Right Power telemetry.
- Printed predict-observe-verify sheet from this guide.
- POWERING ON safety poster visible at the test area.

## Before students enter

- Verify configuration names and basic direction behavior from Lesson 2.
- Confirm both wheels have clearance through their complete path.
- Mark the robot's intended forward direction.
- Set the rule: every mismatch triggers STOP, DISABLED confirmation, and one-change-at-a-time diagnosis.

**SAFETY BOUNDARY** Students never cross into the robot test area while the OpMode is enabled. STOP and visually confirm DISABLED before anyone approaches.

## Student setup and roles

Assign one primary role to each student. Rotate roles on a second trial if time permits. The safety lead may stop the activity at any time.

| STUDENT | ROLE | RESPONSIBILITY |
|---|---|---|
| 1 | **Safety lead** | Runs the POWERING ON and approach checkpoints. |
| 2 | **Driver Station** | Controls INIT, PLAY, and STOP. |
| 3 | **Driver** | Applies one gentle command at a time. |
| 4 | **Left observer** | Reports left-wheel direction and relative speed. |
| 5 | **Right observer** | Reports right-wheel direction and relative speed. |
| 6 | **Telemetry reader** | Reads commanded Left Power and Right Power. |
| 7 | **Recorder** | Captures prediction, command, observation, and result. |
| 8 | **Debugger** | Chooses the next single check after a mismatch. |

### Ground rules

- Predict before testing; record evidence before proposing a correction.
- Only the Driver Station operator touches INIT, PLAY, or STOP.
- Any student may call STOP for unexpected motion, instability, heat, noise, or an unclear boundary.
- Change only one variable between tests so the evidence remains interpretable.

### POWERING ON gate

**SAY AND DO ALL SIX STEPS** 1. Secure the robot. 2. Confirm DISABLED. 3. Look around. 4. Make sure everyone is clear. 5. Call POWERING ON! and wait. 6. Visually confirm clear, then enable/PLAY.

**After every test: press STOP, visually confirm DISABLED, and only then permit anyone to approach the robot.**

# 25-minute teacher walkthrough

## 1. Decode setPower() | 4 minutes

| | |
|---|---|
| **TEACHER** | *“Display leftMotor.setPower(leftPower); and ask students to identify the object, method, and argument.”* |

| | |
|---|---|
| **STUDENT** | *“Students label leftMotor as object, setPower as method, and leftPower as argument/current requested value.”* |

| | |
|---|---|
| **ASK** | *“Which part contains the value being sent?”* |

**LISTEN FOR** The argument inside parentheses: leftPower.

**TEACHING NOTE** Repeat with rightMotor so students see the same pattern applied to a different object.

## 2. Predict from sign and magnitude | 5 minutes

| | |
|---|---|
| **TEACHER** | *“Reveal one left/right pair at a time. Require a written prediction before any live command.”* |

| | |
|---|---|
| **STUDENT** | *“Students use sign for commanded direction and magnitude for relative strength, then predict straight, backward, stop, spin, or curve.”* |

| | |
|---|---|
| **ASK** | *“What should +0.3 and -0.3 do?”* |

**LISTEN FOR** The sides are commanded in opposite directions, producing an in-place turn/spin if the drivetrain is configured correctly.

**TEACHING NOTE** Predictions assume both positive commands correspond to the team's physical forward convention.

## 3. Test one side at a time | 6 minutes

| | |
|---|---|
| **TEACHER** | *“Run the full POWERING ON sequence, press PLAY, move the left stick gently, center it, then move the right stick gently.”* |

| | |
|---|---|
| **STUDENT** | *“Observers report from outside the boundary while telemetry reader reads the commanded values. Recorder compares each to the prediction.”* |

| | |
|---|---|
| **ASK** | *“Why start with one side?”* |

**LISTEN FOR** It isolates the relationship among one stick, one variable, one setPower call, one telemetry value, and one wheel.

**TEACHING NOTE** If anything is unexpected, press STOP before discussing causes.

## 4. Test combined commands | 6 minutes

| | |
|---|---|
| **TEACHER** | *"Use gentle straight, backward, spin, and curve inputs. Pause at center between cases."* |

| | |
|---|---|
| **STUDENT** | *"For each case, students say prediction, read telemetry, observe both wheels, and mark match or mismatch."* |

| | |
|---|---|
| **ASK** | *"What does telemetry prove?"* |

**LISTEN FOR** It shows the values commanded by the program; it does not independently measure wheel speed or direction.

**TEACHING NOTE** Keep tests short. Heat, noise, or instability triggers immediate STOP.

## 5. Diagnose safely | 4 minutes

| | |
|---|---|
| **TEACHER** | *"Press STOP and confirm DISABLED before the team approaches. Choose only one evidence-based check."* |

| | |
|---|---|
| **STUDENT** | *"The debugger selects configuration name, direction setting, wiring, mechanical binding, or input/telemetry as the next check."* |

| | |
|---|---|
| **ASK** | *"Why change only one thing?"* |

**LISTEN FOR** A single controlled change lets the team know which correction affected the result.

**TEACHING NOTE** Repeat the entire POWERING ON sequence before every retest.

## Predict - command - observe - verify

Student/team: ________________________________ Date: ______________ Robot: __________________

| LEFT / RIGHT | PREDICTION | TELEMETRY + OBSERVATION | MATCH? / NEXT CHECK |
|---|---|---|---|
| **0.0 / 0.0** | | | |
| **+0.3 / +0.3** | | | |
| **-0.3 / -0.3** | | | |
| **+0.3 / -0.3** | | | |
| **+0.5 / +0.2** | | | |
| **Student test: ____ / ____** | | | |

**STUDENT EVIDENCE RULE** Write what was commanded and what was observed. Do not write a correction until STOP has been pressed and DISABLED has been confirmed.

### Safety sign-off

Before live testing, the safety lead confirms: robot secure [ ] DISABLED [ ] boundary clear [ ] POWERING ON called and wait completed [ ]

**After testing: STOP pressed [ ] DISABLED visually confirmed [ ] Safety lead initials: __________**

**COMPLETION CRITERIA** The student uses lesson vocabulary, predicts before testing, records evidence, explains one result from the code or math, and follows STOP/DISABLED rules before approaching the robot.

## Debrief

| ASK | LISTEN FOR |
|---|---|
| **What does setPower(0.0) request?** | No motor output, while the OpMode may continue running. |
| **What do sign and magnitude represent?** | Sign is commanded direction relative to configuration; magnitude is requested strength. |
| **Why can telemetry and wheel motion disagree?** | Telemetry shows the command, while mapping, direction configuration, wiring, or mechanics determine physical response. |
| **What is the response to any mismatch?** | STOP, confirm DISABLED, then diagnose one item at a time. |

## Troubleshooting without guessing

| OBSERVATION | NEXT CHECK AFTER STOP/DISABLED |
|---|---|
| **Telemetry correct; wheel direction wrong** | Check motor Direction after STOP/DISABLED. |
| **Telemetry correct; no wheel motion** | Check mapping, wiring, hub connection, battery, and mechanical binding. |
| **Left telemetry changes; right wheel moves** | Recheck configuration-name-to-port mapping. |
| **Robot shifts on stand** | STOP immediately; improve support and clearance before any retest. |

## Technical notes for the teacher

- setPower(double) receives the program's requested motor output, normally within -1.0 to +1.0.
- Sign is relative to configured motor direction; mounting determines physical wheel motion.
- Equal values produce straight motion only when directions and mechanics are aligned.
- Telemetry reports commanded values, not an independent wheel-speed measurement.
- The runtime disables outputs after STOP, but the team must still visually confirm DISABLED before approach.

**EXIT TICKET** Telemetry shows +0.3 for a wheel that turns physically backward. List the first two facts you would verify after STOP and DISABLED.